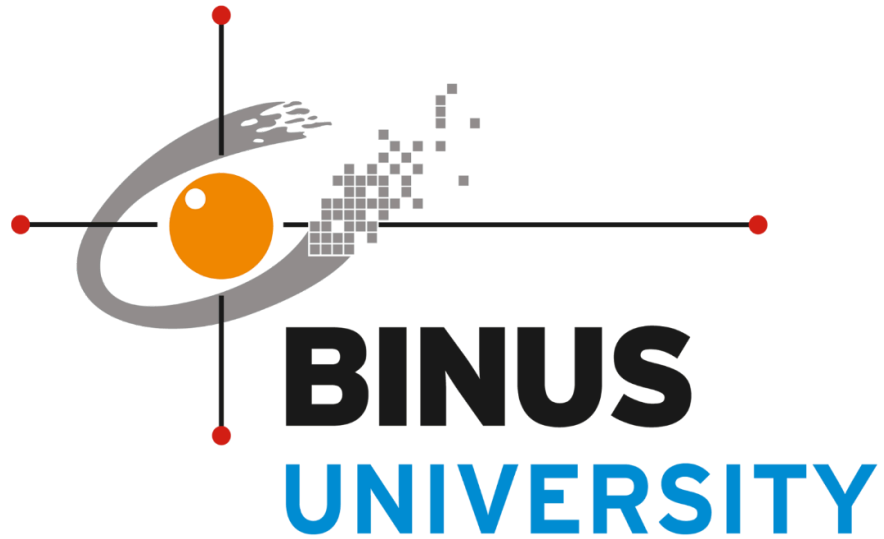


P MINUM (WATER INTAKE TRACKER)



GROUP 3

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CHAPTER I

INTRODUCTION

1.1 Background

Water is a vital nutrient essential to nearly every physiological process in the human body. It acts as a medium for biochemical reactions, regulates body temperature, maintains blood volume, transports nutrients, and facilitates waste elimination. Despite its irreplaceable role in health and well-being, many individuals, particularly students and working adults, fail to consume sufficient amounts of water daily. This often results from busy routines, lack of awareness, or the absence of immediate symptoms, leading to chronic dehydration.

Even mild dehydration can impair physical performance and cognitive functions such as memory, concentration, and reaction time. In academic settings, university students are especially vulnerable. A study at Universitas Gadjah Mada found high levels of dehydration among students, largely due to inconsistent drinking habits and the consumption of diuretic beverages like coffee and energy drinks. Long-term dehydration in this population has been associated with psychological distress and diminished academic performance.

The consequences of inadequate fluid intake range from mild symptoms like fatigue and headaches to severe outcomes including kidney stones, urinary tract infections, cardiovascular issues, and even death. Several reported cases in Indonesia highlight the severity of this issue. For example, a university student who died after collapsing from dehydration during a physical initiation event. These cases underscore the need for increased awareness and proactive hydration habits, especially among youth who may not recognize the early signs of dehydration.

Conversely, overhydration or water intoxication, while less common, poses serious health risks such as electrolyte imbalance, brain swelling, and hyponatremia, a potentially life-threatening condition caused by diluted sodium levels in the blood. This highlights the importance of balanced hydration tailored to individual needs, rather than simply encouraging increased water intake.

To counter these issues, an application is built to track user's daily water intake. Our study aims to develop an application to track user's daily water intake, create a machine learning model that helps classify the weight of a user's daily activity, and analyze performance of our developed model in doing the classification task.

CHAPTER II

METHODOLOGY

2.1 Data Collection

To ensure comprehensive training and evaluation of the model, synthetic data is created to simulate diverse user behaviors which includes varied daily routines, habits, and activities. It allows the system to account for a wider range of scenarios. The synthetic data helps to build a robust model that can make accurate predictions when real user data are limited or unavailable. In total, there are around 300 rows of synthetic data generated to be used to train the model.

2.2 Data Preprocessing

In data processing, data is cleaned, transformed, normalized raw user input into usable features for machine learning models. The processing is important to ensure that the data used for the system is suitable for making an accurate prediction.

1. Tokenization and Lemmatization: The texts are splitted into words and are reduced to its based form (e.g running -> run).
2. Feature Engineering: For the feature extraction, we use TF-IDF which converts the texts to numerical features.
3. Normalization: Text normalization is applied in the data preprocessing by doing lemmatization, making the words into lowercase, and removing the stop words.

2.3 Model Training and Testing

For the classification model, we decided to use a multinomial naive bayes model due to the lack of dataset for training. The dataset is first splitted into 80% training and 20% testing using a stratified approach to preserve the class distribution. A pipeline is then created to combine the TF-IDF feature extraction and the selected model. Next, we fine-tune the hyperparameters by using GridSearchCV with 5 fold cross validation. The hyperparameters that are fine-tuned are: max_df, min_df, ngram_range, use_idf, smooth_idf, sublinear_tf, and alpha.

CHAPTER III IMPLEMENTATION

3.1 Technologies Used

The machine learning model for activity classification was developed in a Python environment using several key libraries:

1. Pandas: Utilized for creating and managing the synthetic dataset used to train the model.
2. Scikit-learn: A comprehensive library for machine learning in Python.
 - a. TfidfVectorizer for converting textual activity descriptions into a matrix of TF-IDF features.
 - b. MultinomialNB as the classification algorithm to predict activity intensity.
 - c. Pipeline to streamline the workflow of text vectorization and model training.
 - d. GridSearchCV for systematic hyperparameter tuning to find the optimal model configuration.
 - e. train_test_split for partitioning the data into training and testing sets for robust evaluation.
 - f. confusion_matrix and ConfusionMatrixDisplay for model performance analysis.
3. NLTK (Natural Language Toolkit): Employed for text preprocessing, specifically using the WordNetLemmatizer to normalize words to their base form.
4. Joblib: Used for serializing and saving the trained model pipeline for later use in the application.
5. Matplotlib: Used for visualizing the confusion matrix to evaluate model performance.

3.2 Formulas

Our application utilizes user's bio information such as weight, health condition, and weight of daily activity to calculate the daily water intake user's are expected to reach.

3.2.1. The Holliday-Segar Method (Healthy People)

This method calculates maintenance fluid needs based on caloric expenditure, which is closely correlated with weight in a predictable, non-linear fashion [19]. It is considered a gold standard in pediatrics and offers superior accuracy for children and adolescents compared to simple linear weight-based formulas designed for adults.

The method, often simplified as the "4-2-1 Rule" for hourly rates or the "100-50-20 Rule" for daily rates, is calculated as follows:

- For the first 10 kg of body weight: 100 mL per kg per day.
- For the next 10 kg of body weight (from 11 kg to 20 kg): 50 mL per kg per day.
- For each kilogram of body weight above 20 kg: 20 mL per kg per day.

The formula can be expressed as:

$$\text{Daily Intake (mL)} = (100 \times \min(\text{Weight (in kg)}, 10)) + (50 \times \min(\max(0, \text{Weight (in kg)} - 10), 10)) + (20 \times \max(0, \text{Weight (in kg)} - 20))$$

3.2.2. People with Diabetes: base + symptomatic increase

Although no specific global equation exists, clinical guidelines for diabetic hydration recommend an increase of 500 mL/day over baseline due to polyuria. This aligns with typical advice across ADA and CDC hydration guidance.

Equation for people with diabetes:

$$\text{Intake} = (30 \times \text{weight}) + 500$$

3.2.3. Pregnant Women

The *EPA Exposure Factors Handbook* states pregnant women average about 300 mL/day more intake – translating approximate daily needs to 2.700 mL. The EFSA confirms a +300mL/day increase during pregnancy.

Equation for Pregnant Women:

$$\text{Intake} = (30 \times \text{weight}) + 300$$

3.2.4. Breastfeeding Mothers

The *EPA Handbook* notes lactating women consume significantly more (approximately 26 mL/kg vs approximately 15 mL/kg baseline). Institute of Medicine and Mayo brochure estimates breast milk water needs lead to roughly +700 mL/day above baseline.

Equation for Breastfeeding Mothers

$$\text{Intake} = (30 \times \text{weight}) + 700$$

3.2.5. Physical Activity and Exertion Adjustment

Physical activity is one of the most significant variables affecting daily fluid requirements. As exercise intensity and duration increase, the body loses substantial water and electrolytes through sweat to regulate temperature. Hence, the application uses the Simple Additive Model, which adds about 350 mL of water for every 30 minutes of moderate exercise. Since the activity level is categorized by weight (light, medium, and heavy), the corresponding estimated exercise durations are 30, 60, and 90 minutes, respectively.

Equation for Physical Activity and Exertion Adjustment

$$\text{Activity Adjustment (oz)} = \frac{\text{Total Exercise Duration (min)}}{30} \times 12$$

3.3 Deployment Structure

The trained machine learning pipeline, including the text vectorizer and optimized classifier is saved as a single file (activity_pipeline.pkl) using *Joblib*. This file can be easily loaded into the main application, making it simple to integrate and ensuring consistent results between training and real-world use.

3.4 Application Development Integration

FlutterFlow is utilized in order to make the user interface for the application. A screenshot of the application page that integrates the machine learning model is attached in the provided link: 

3.5 Machine Learning Integration

The *activity_pipeline.pkl* model is designed for seamless integration into the final Android application. The integration workflow is as follows:

1. The user inputs a description of their daily activities in a text field within the app (e.g., "I went for a 3-mile run, then had a protein shake").
2. This text is sent to the application's backend server, where the saved pipeline is loaded.
3. The pipeline's *.predict()* method is called with the user's text. The pipeline automatically handles the entire preprocessing and classification task.
4. The model returns a predicted activity level: "light", "medium", or "heavy".
5. This classification output is then used by the application to adjust the user's personalized daily water intake recommendation, fulfilling a core objective of the system.

CHAPTER IV EVALUATION AND RESULTS

4.1 Model Performance Evaluation

The model's performance was evaluated using two primary methods based on the code provided:

1. Cross-Validation Accuracy: To find the best model and estimate its ability to generalize, a 5-fold cross-validation strategy was used across the entire dataset. The best-performing model configuration achieved a mean cross-validated accuracy of 78.6%. This score represents the average performance on each fold when it was used as a test set during the grid search process.
2. Overall Model Accuracy: After the grid search identified the best hyperparameters, the final pipeline was trained on the entire dataset. When this final model was used to make predictions on the same dataset, it achieved an overall accuracy of 83%. This metric indicates how well the final model learned to classify the provided data.

4.2 Hyperparameter Tuning Results

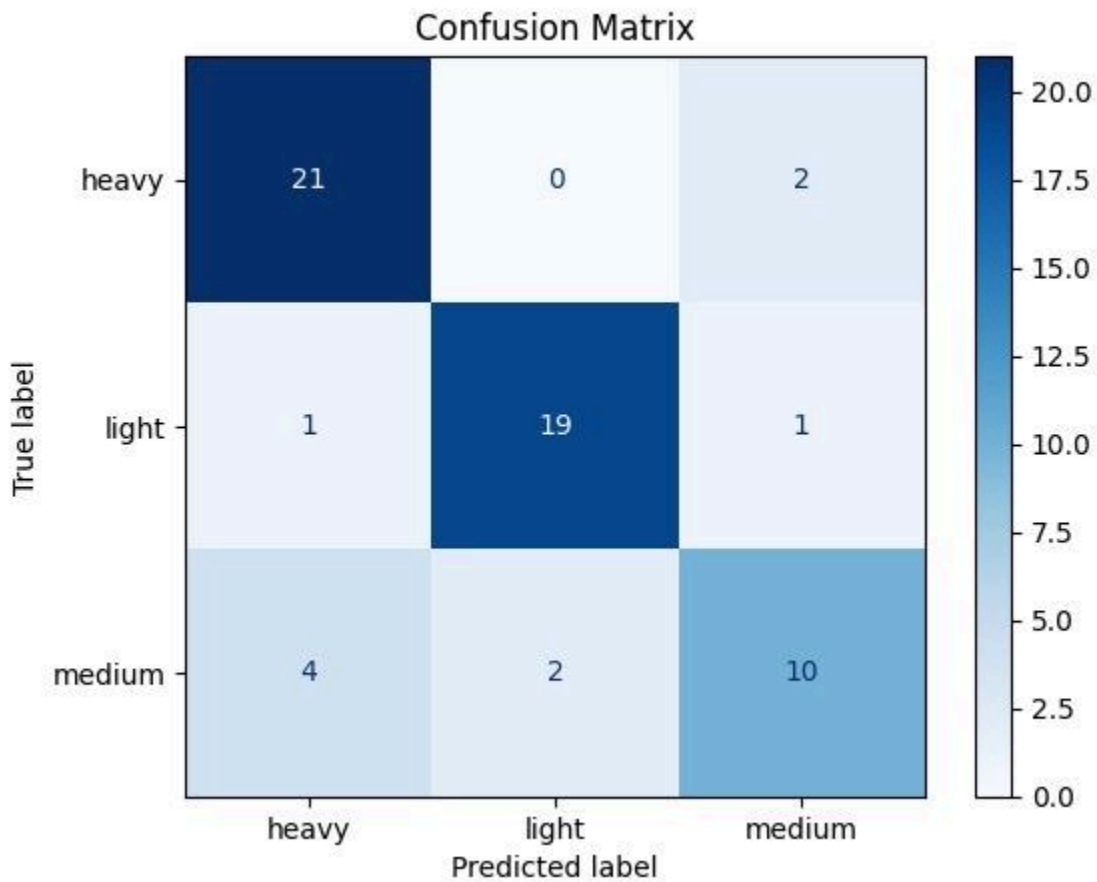
A *GridSearchCV* was employed to find the optimal combination of parameters for the *TfidfVectorizer* and the *MultinomialNB* classifier. The best parameters identified through this process are as follows:

- `model__alpha`: 5
- `tfidf__max_df`: 0.5 (ignores terms that appear in more than 50% of the documents)
- `tfidf__min_df`: 1 (ignores terms that appear in less than 2 documents)
- `tfidf__ngram_range`: (1, 1) (considers only single words, not phrases)
- `tfidf__smooth_idf`: True
- `tfidf__sublinear_tf`: True
- `tfidf__use_idf`: False

This specific combination of parameters was used to train the final model that is saved for application deployment.

4.3 Confusion Matrix Analysis

A confusion matrix was generated to provide a detailed, visual breakdown of the final model's classification performance for each intensity level. The matrix allows for a clear analysis of not only what the model predicted correctly but also where it made errors.



The results from the matrix are broken down by each true class:

1. Heavy Activity (23 total instances):

- Correctly Predicted: 21 instances were accurately classified as "heavy".
- Misclassified: 0 instances were incorrectly labeled as "light", and 2 were labeled as "medium".
- Class Accuracy (Recall): The accuracy for the "heavy" class is

$$\frac{21}{23} = 91.3\%$$

2. Light Activity (21 total instances):

- Correctly Predicted: 19 instances were accurately classified as "light".
- Misclassified: 0 instances were incorrectly labeled as "heavy", and 1 were labeled as "medium".
- Class Accuracy (Recall): The accuracy for the "light" class is

$$\frac{19}{21} = 91.3\%$$

3. Medium Activity (16 total instances):

- Correctly Predicted: 10 instances were accurately classified as "medium".
- Misclassified: 4 instances were incorrectly labeled as "heavy", and 2 were labeled as "light".
- Class Accuracy (Recall): The accuracy for the "medium" class is

$$\frac{10}{16} = 62.5\%$$

4.4 Classification Report Analysis

	precision	recall	f1-score	support
heavy	0.81	0.91	0.86	23
light	0.90	0.90	0.90	21
medium	0.77	0.62	0.69	16
accuracy			0.83	60
macro avg	0.83	0.81	0.82	60
weighted avg	0.83	0.83	0.83	60

Precision:

The model demonstrates strong precision across all three activity classes. The "light" class achieves the highest precision at **0.90**, followed by "heavy" at **0.81** and "medium" at **0.77**. This indicates that when the model predicts "light," it is correct 90% of the time. Although "medium" has slightly lower precision, it still shows reasonable accuracy in positive predictions.

Recall:

Recall is highest for the "heavy" and "light" categories, both at **0.91** and **0.90** respectively, meaning the model successfully identifies nearly all actual instances of these classes. However, recall for the "medium" class drops to **0.62**, suggesting that the model struggles to correctly capture all actual "medium" instances, potentially misclassifying them as "heavy" or "light."

F1-Score:

The F1-score, which balances precision and recall, is highest for "light" (**0.90**) and "heavy" (**0.86**), reflecting a well-rounded performance for those classes. The "medium"

class has a noticeably lower F1-score of **0.69**, indicating room for improvement in predicting this category.

Overall, the model achieves an overall accuracy of 0.83, indicating that it correctly classifies 83% of all instances. A macro F1-score of 0.82 suggests that the model performs consistently across all classes, although its performance is slightly impacted by lower precision or recall in the "medium" class. Meanwhile, the weighted F1-score of 0.83 accounts for class imbalance and further confirms that the model maintains strong overall performance, particularly in scenarios where some classes are more prevalent than others.

4.5 Results and Discussion

The intelligent hydration application successfully integrates a personalized approach to manage user water intake, addressing a widespread health issue, particularly among students and workers. Using data-driven insights and machine learning, the application is capable of tracking user progress to improve their hydration habits.

By using synthetic data for model training, the system uses a robust dataset that simulates various user behaviours and environmental conditions which compensates for the potential limitations of real user data. It creates a higher accuracy for the model as it is learning from various different scenarios.

To predict the weight of daily activity, Multinomial Naive Bayes classifier is used to provide insights to user activity levels which are used to adjust personalized hydration recommendations. The model achieved an accuracy of 83%, which is already good enough considering the dataset used to train the model is only around 300 rows. In addition, the model can be further improved by using a richer dataset.

Lastly, while the app offers gamification features, further studies could investigate the effectiveness of these features in promoting long-term user engagement. The feedback mechanisms(eg. badges and progress tracking) seem to be effective to encourage users.

CONCLUSION

In conclusion, our application and machine learning model demonstrate that personalized hydration recommendations through a mobile app are both feasible and effective. By integrating synthetic data, user profiles, and optional wearable device input, the app can deliver tailored hydration reminders and monitor individual progress to support better hydration habits. While the Multinomial Naive Bayes model provides a solid baseline, future improvements could include adopting more advanced machine learning algorithms and incorporating a larger, more diverse dataset. Additionally, enhancing the app's features, such as gamification elements and personalized tips, will be essential for maintaining user engagement and promoting sustained behavior change.

This application is a flexible and scalable solution to the dehydration problem that has been happening lately. It gives users personalized hydration reminders that fit easily into their daily routines. In the future, improvements based on user feedback would improve the app better on users' hydration needs.

EVALUATION STATEMENT

Steven Jonatan	UI/UX Design: I designed the app's UI/UX, prioritizing ease of use and clear, actionable presentation of machine learning feedback (e.g., activity classification, water intake recommendations). Design Refinements: Using Flutterflow, I developed wireframes and prototypes, refining UI components for ML results like dynamic hydration suggestions, and enhanced the UI based on feedback. Feedback Integration: I listened to feedback from the team and made design changes to improve the app's usability.
Kent Putra Lamiran	UI/UX Design: I helped create a design that worked well on mobile devices and was easy for users to interact with. I made sure the layout was clean and simple for users to follow. User Experience: I worked on mapping out how users would move through the app, making sure they could use all the features, including the Machine Learning input. Prototyping and Testing: I helped build prototypes to test the design with potential users and adjusted the design based on what they liked or didn't like.
Rosaline Imely Gosang	App Development: Designed and maintained app's backend logic, including user authentication for secure interface-database interaction. Integrating Machine Learning: Developed ML model for personalized hydration, integrating it into the app to classify daily activities and predict optimal water intake. Managed data preprocessing,

	UI integration, and on-device optimization. Video Production: I produced and edited promotional videos showcasing the app's functionality and hydration benefits to attract users, ensuring a professional presentation that highlighted key features.
Joe Kenryoku Fendy	App Development: I worked on the backend of the app. I made sure that all the data, like user information and hydration logs, were stored securely and could be accessed easily. Database and API Integration: Managed app database for real-time data interaction and integrated it with the ML pipeline to store, process, and pass user activity inputs to the model for real-time classification and hydration recommendations. Machine Learning Development: Responsible for developing the machine learning model (data preprocessing, model training and testing).
Kevin Nathaniel	Machine Learning Model Integration: I integrated the machine learning model into the app, which predicts daily water intake based on the user's daily activity. Prediction and Data Flow: I made sure that when users entered their activities, the app could predict their hydration needs and provide personalized recommendations. Optimization: I worked on improving the app's speed and performance, making sure it used as little battery as possible while still providing accurate predictions.

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